Paper questions HL E.1 [28 marks]

1. [Maximum mark: 1]

An alpha particle ($^{4}_{2}\mathrm{He}$) of initial energy 5.5 MeV moves towards the centre of a nucleus of gold-197 ($^{197}_{79}\mathrm{Au}$).

What is the distance of closest approach of the alpha particle?

A. 1.0×10^{-13} m

B. 4.1×10^{-14} m

C. 2.1×10^{-14} m

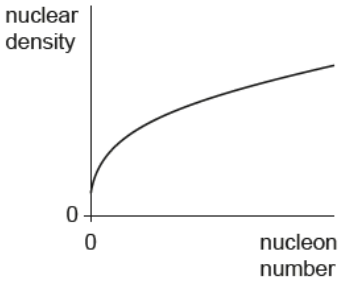
D. 6.6×10^{-33} m [1]

Markscheme
B

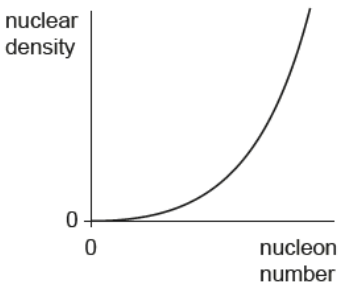
2. [Maximum mark: 1]

What is the variation with nucleon number of the density of nuclei?

A.

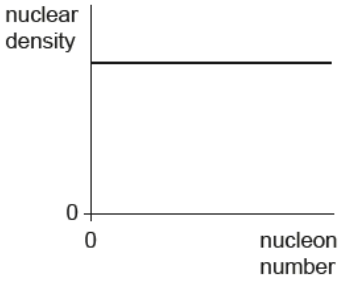


B.

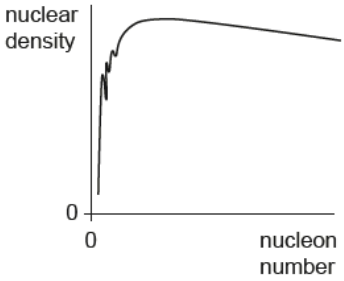


[1]

C.



D.



Markscheme
C

3. [Maximum mark: 1]

In the Geiger–Marsden–Rutherford experiment, alpha particles were accelerated through a potential difference of 7.7 MV and incident on a thin foil of gold. The proton number of gold is 79.

What is the distance of closest approach?

A. $\frac{158ke}{7.7\times10^6}$

B. $\frac{158ke^2}{7.7\times10^6}$

C. $\frac{79ke}{7.7\times10^6}$

D. $\frac{79ke^2}{7.7\times10^6}$ [1]

Markscheme
C

4. [Maximum mark: 1]

The nucleus of an element X has radius R and density ρ.

The nucleon number of an element Y is 8 times that of X.

What is the radius and density of the nucleus of Y?

	Radius of nucleus of Y	Density of nucleus of Y
A.	$8R$	8ρ
B.	$2R$	8ρ
C.	$8R$	ρ
D.	$2R$	ρ

[1]

Markscheme
D

5. [Maximum mark: 8]

(a) Rutherford modelled the scattering of alpha particles by a nucleus of gold. However, when the initial alpha particle energy is greater than a certain value E_0, the fraction of alpha particles scattered at angles close to 180° does not follow Rutherford's model.

Outline

(a.i) why this deviation from the model occurs, and [1]

Markscheme
«at closest approach alpha is within the range of the» strong nuclear interaction/force «of the gold nucleus»

(a.ii) how E_0 can be used to estimate the nuclear radius of gold. [2]

Markscheme
Recognizes that nuclear radius is «approximately» the distance of closest approach Explains calculation of r from energy conservation, e.g. $E_0 = k\frac{Qq}{r}$ with symbols explained or description of initial KE of alpha particle being converted to electrical potential energy. *MP1 can be implied if the answer discusses the distance of closest approach.*

Use of r in an algebraic expression isn't sufficient for MP1.

Gold-198 (Au-198) is a radioactive nuclide of gold. Au-198 decays into stable mercury-198 (Hg-198) with an emission of a beta particle followed by one or two gamma photons.

The half-life of Au-198 is 2.69 days.

(b.i) Calculate, in s^{-1}, the decay constant of Au-198. [1]

Markscheme

$\frac{\ln 2}{2.69\times24\times3600} = 2.98 \times 10^{-6}\,s^{-1}$

Unit is not required.

(b.ii) A sample contains 5.0 mg of pure Au-198. Determine the mass of Hg-198 present in the sample after one week. [2]

Markscheme

Undecayed mass $= 5.0 \times e^{-2.98\times10^{-6}\times7\times24\times3600}$ «=0.82 mg»

OR

$5.0 \times \left(\frac{1}{2}\right)^{\frac{7.0}{2.69}}$ «=0.82 mg»

Mass of Hg-198 = «5.0 – 0.82=»4.2 mg

Unit is not required.

Check the steps in the calculation.

If the unit is missing in the answer, check the calculation to see whether the candidate is working in mg or kg and ensure that the power of ten is correct. If any other unit is given, the power of ten and the prefix must match e.g. 4.2×10^3 μg.

Gamma photons of the following frequencies are emitted in the decay of Au-198:

Frequency / 10^{20} Hz
1.00
1.63
2.63

(c) Explain how this observation provides evidence that nuclear energy levels are discrete. [2]

Markscheme

«$E = hf$ hence» observed gamma photons have fixed / discrete energies

OR

Use of $E = hf$ to calculate at least one photon energy

Gamma energy equals the difference between nuclear energy levels

OR

Gamma photons are emitted when the nucleus undergoes transitions between its energy levels

Energies are

6.63×10^{-14} J **OR** 0.41 MeV

1.08×10^{-13} J **OR** 0.68 MeV

1.74×10^{-13} J **OR** 1.1 MeV

Do not award MP2 if transitions of electrons are discussed.

6. [Maximum mark: 1]
The radius of a nucleus of a particular nuclide X is 4.8 fm.

What is the nucleon number of X?

A. 48

B. 64

C. 128

D. 148 [1]

Markscheme
B

7. [Maximum mark: 1]
When alpha particle scattering experiments were carried out with high-energy alpha particles, deviations from Geiger-Marsden-Rutherford scattering were observed.

What was deduced as a result of this observation?

A. The size of the alpha particle

B. The size of the nucleus

C. The nature of the electrostatic field inside the nucleus

D. The nature of the weak nuclear force within the nucleus [1]

Markscheme

B

8. [Maximum mark: 7]

(a.i) In the Bohr model for the hydrogen atom orbital angular momentum is quantized $\left(mvr = n\frac{h}{2\pi}\right)$. Outline the effect this has on the electron energy. [1]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

The energy of the electron is discrete/quantized

(a.ii) For the hydrogen atom, the orbit radius of the electron in the $n = 2$ state is four times larger than the orbit radius in the $n = 1$ state. Determine the ratio $\frac{v_2}{v_1}$ of the electron speed in the $n = 2$ state to the speed in the $n = 1$. [2]

Markscheme

ALTERNATE 1

$v \propto \frac{1}{\sqrt{r}}$ ✓

Leading to $\frac{v_2}{v_1} = \frac{1}{\sqrt{4}} = \frac{1}{2}$ ✓

ALTERNATE 2

$E_k = -E_T = \frac{13.6}{n^2}$ «eV» ✓ $\frac{\frac{1}{2}mv_2^2}{\frac{1}{2}mv_1^2} = \frac{\frac{13.6}{4}}{\frac{13.6}{1}} = \frac{1}{4}$ ✓

Leading to $\frac{v_2}{v_1} = \frac{1}{\sqrt{4}} = \frac{1}{2}$ ✓

ALTERNATE 3

Use of quantization condition: $\frac{mv_2r_2}{mv_1r_1} = \frac{2 \times h}{1 \times h}$ ✓

$\frac{v_2}{v_1} = 2 \times \frac{1}{4} = \frac{1}{2}$ ✓

(b) Compare and contrast the Bohr prediction for the radius of an electron orbit in hydrogen to that of the description of the electron in terms of a wave function. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

The Bohr model predicts a specific orbit radius for each level/state ✓

The wavefunction assigns a probability «distribution» for the electron to be at various distances from the nucleus «for each level/state»
OR
the Bohr radius is the most likely electron distance «from the nucleus» ✓

(c) Determine, using the uncertainty principle, the minimum kinetic energy of a neutron inside a nucleus of radius 3×10^{-15} m. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

$\Delta p =$« $\frac{h}{4\pi\Delta x}$ =» $\frac{6.63 \times 10^{-34}}{4\pi \times 3 \times 10^{-15}}$ «= 1.76×10^{-20}» «Ns»

$E_k =$« $\frac{(\Delta p)^2}{2m_n} = \frac{\left(1.76 \times 10^{-20}\right)^2}{2 \times 1.67 \times 10^{-27}}$ =» 9.2×10^{-14} ***OR*** 9.3×10^{-14} «J» *or* 0.58 «MeV»

9. [Maximum mark: 1]

The results obtained from Geiger-Marsden-Rutherford scattering experiments provide evidence for the existence of the nucleus in atoms. Deviations from Geiger-Marsden-Rutherford scattering…

I. provide evidence for the existence of the strong nuclear force.

II. allow for an estimation of the radius of the nucleus.

III. occur when alpha particles are incident at low energies.

Which statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III [1]

Markscheme

A

10. [Maximum mark: 1]

In the Bohr model for the hydrogen atom, what is $\frac{\text{angular momentum of first excited state}}{\text{angular momentum of ground state}}$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 2

D. 4 [1]

Markscheme

C

11. [Maximum mark: 1]

In the Bohr model for hydrogen, the radius of the electron orbit in the n = 2 state is four times that of the radius in the n = 1 state.

What is $\frac{\text{speed of the electron in the n} = 2 \text{ state}}{\text{speed of the electron in the n} = 1 \text{ state}}$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 2

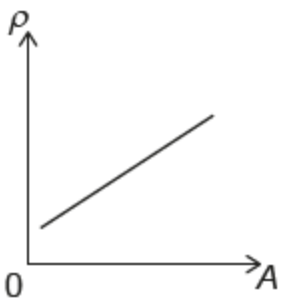
D. 4 [1]

Markscheme
B

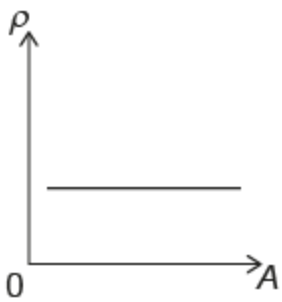
12. [Maximum mark: 1]

What is the variation of nuclear density ρ with nucleon number A?

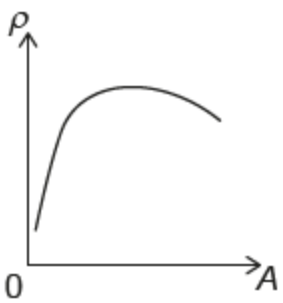
A.



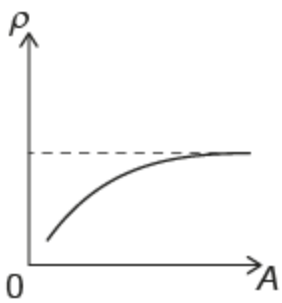
B.



C.



D.



[1]

Markscheme
B

13. [Maximum mark: 1]

The nucleus of the isotope hydrogen-2 has a radius R and a density ρ.

What are the approximate radius and density of a nucleus of oxygen-16?

	Oxygen-16 radius	Oxygen-16 density
A.	$2R$	ρ
B.	$2R$	2ρ
C.	$8R$	ρ
D.	$8R$	2ρ

[1]

Markscheme

A

14. [Maximum mark: 1]

Element X has a nucleon number A_{X} and a nuclear density ρ_{X}. Element Y has a nucleon number of $2A_{\mathrm{X}}$. What is an estimate of the nuclear density of element Y?

A. $\frac{1}{2}\rho_{\mathrm{X}}$

B. ρ_{X}

C. $2\rho_{\mathrm{X}}$

D. $8\rho_{\mathrm{X}}$

[1]

Markscheme

B

15. [Maximum mark: 1]

The diameter of a nucleus of a particular nuclide X is $12\ \text{fm}$. What is the nucleon number of X?

A. 5

B. 10

C. 125

D. 155 [1]

Markscheme
C

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